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Question

Some astronomers searching for extraterrestrial life have proposed that atmospheric NH_3 (ammonia) can serve as a biosignature gas—an indication that a planet harbors life. Jingcheng Huang, Sara Seager, and colleagues evaluated this possibility, finding that on rocky planets, atmospheric NH_3 likely couldn't reach detectably high levels in the absence of biological activity. But the team also found that on so-called mini-Neptunes—gas planets smaller than Neptune but with atmospheres similar to Neptune's—atmospheric pressure and temperature can be high enough to produce atmospheric NH_3 .

Based on the text, Huang, Seager, and colleagues would most likely agree with which statement about atmospheric NH_3 ?

Answer

- A. Its presence is more likely to indicate that a planet is a mini-Neptune than that the planet is a rocky planet that could support life.
- B. Its absence from a planet that's not a mini-Neptune indicates that the planet probably doesn't have life.
- C. It should be treated as a biosignature gas if detected in the atmosphere of a rocky planet but not if detected in the atmosphere of a mini-Neptune.
- D. It doesn't reliably reach high enough concentrations in the atmospheres of rocky planets or mini-Neptunes to be treated as a biosignature gas.